Modeling Emotion Recognition Under Uncertainty in Internet Comments

Yana Zlatanova
yana.zlatanova.gold@gmail.com

André Plancha
andre.plancha@hotmail.com

Vérane Flaujac
vflaujac@gmail.com

Abstract

Emotion recognition in text is a crucial challenge in natural language processing, particularly when dealing with noisy, subjective annotations such as those found in internet comments. We explore uncertainty-aware emotion classification using the GoEmotions dataset, a large, multi-label corpus of Reddit comments labeled with 27 nuanced emotion categories. We fine-tune DistilBERT models using multiple emotion taxonomies and label aggregation strategies that preserve annotator disagreement by converting classification into a multi-label regression problem. Three loss functions – Mean Binary Cross Entropy, SoftRank-approximated nDCG, and Weighted Mean Squared Error – were evaluated to assess their impact on model performance across classification, ranking, and confidence-based metrics. Results show that despite dataset noise and imbalance, DistilBERT can learn meaningful emotion representations and effectively capture emotion uncertainty.

Keywords

Emotion Recognition, NLP, GoEmotions, Internet Comments, Machine Learning, HuggingFace, DistillBERT, Evaluation Metrics

1 Introduction

As the flavor of speech that determines the meaning beyond the literal translation, emotions are essential to human communication. Due to helping machines better understand human text and speech, emotion recognition is a crucial subfield of natural language processing (NLP). This is especially helpful in chatbots, customer feedback analysis, and mental health monitoring[16]. However, there are difficulties in recognizing emotions. Conventional machine learning models are ineffective and unable to handle the complexity of human language because they require a great deal of feature engineering. However, even though deep neural networks perform better, they usually need a lot of labeled training data, which is hard to obtain in a real world context [2].

One of the ways to address this challenge is with transfer learning, by using pre-trained transformer-based natural language models such as BERT, RoBERTa, DistilBERT, XLNet, etc. These models are trained on massive amounts of data and are capable of learning complex linguistic structures and contextual relationships. Their knowledge can be transferred by fine-tuning them on a specialized emotion-labeled dataset for effective emotion classification with no training from scratch [2].

In order to address the following research questions, this study investigated emotion classification in text using DistilBERT-based models [18], a smaller and faster version of BERT.

- What are the challenges in emotion recognition with a human annotated dataset?
- Can fine-tuned models reliably classify emotions from Reddit comments, despite annotator disagreement?
- How does emotion classification performance differ across taxonomies?
- What is the effect of different loss functions on the fine-tuning performance of DistilBERT?

These models were fine-tuned on the GoEmotions dataset [7], which contains approximately 58,000 Reddit comments annotated with 28 emotion labels.

This report serves as a comprehensive overview of our study, detailing the dataset and our exploration of it, previous research done on both the GoEmotions dataset and in sentiment and emotion analysis the transformations applied and task formulation, the model architectures and performance metrics, and the results of our experiments. In short, it describes the various issues we found with the dataset, how we solved them, including how we decided to incorporate human annotator disagreements, how we decided to model and evaluate the models, and the general conclusions of the study.

2 Dataset

GoEmotions is the largest human-annotated emotion dataset, with multiple labels per comment to ensure quality [7]. This section outlines how the dataset was collected, annotated, and processed, following the original scientific publication by its creators [7].

A key innovation is its 27-emotion taxonomy, illustrated in Figure 2, based on modern psychological research and going beyond Ekman's six basic emotions. The dataset includes English-only Reddit comments from subreddits containing at least 10k comments.

2.1 Annotation Process

To ensure annotation quality, each comment was reviewed by multiple raters [7] who categorized the comment into to emotions. Initially, three annotators assessed each comment. If there was no agreement on at least one emotion label, two additional annotators were assigned. All raters were native English speakers from India

Sample Text	Label(s)
OMG, yep!!! That is the final answer. Thank you so much!	gratitude, approval
I'm not even sure what it is, why do people hate it	confusion
Guilty of doing this tbph	remorse
This caught me off guard for real. I'm actually off my bed laughing	surprise, amusement
I tried to send this to a friend but [NAME] knocked it away.	disappointment

Figure 1: Snippet of the GoEmotions Dataset.

and they were presented with the comments without author or subreddit information [1].

2.2 Taxonomy

It is important for us to know how was the data set collected, put together and cleaned for us to be able to interpret the results correctly.

The 27-category emotion taxonomy of the dataset was inspired by modern psychological research which is far beyond the traditional six basic emotions — joy, anger, fear, sadness, disgust, and surprise — originally proposed by Ekman [7].

Comments containing offensive or adult language were removed, except for vulgar comments, which were kept to help study negative emotions. Comments with offensive content toward minorities were manually removed. Only comments with 3 to 30 tokens (including punctuation) were retained. Various techniques were applied to balance the dataset and reduce emotion overrepresentation. Additionally, personal names and religion terms were masked with [NAME] and [RELIGION] tokens, respectively. Note that raters saw the original, unmasked comments during annotation [7].

3 Literature review

The GoEmotions dataset was introduced by Demszky, D. et al. [7], which outlines the motivation, processes, and tools used to create the dataset we are using, along with experiments showcasing its effectiveness. The GoEmotions dataset was introduced to address the lack of sufficiently large datasets for language-based emotion classification and the limitations of existing emotion taxonomies, which typically use limited emotion taxonomies, such as Ekman's 6 emotions [8]. Demszky, D. et al. claims they created the largest human-annotated dataset of 58k carefully selected Reddit comments, labeled with 27 emotion categories or Neutral, as shown on Figure 2, drawn from popular English subreddits. The dataset stands out for its richer taxonomy, which includes a more diverse range of positive, negative, and ambiguous emotions; in contrast, Ekman's taxonomy includes only one positive emotion (joy).

The paper explains how the dataset was constructed and presented a baseline BERT-based model for emotion prediction, achieving a F_1 of 0.46 over the proposed 27 emotions taxonomy, but performed better with a 0.64 score using an Ekman-style grouping into six emotion categories and 0.69 using a simple sentiment grouping (positive, neutral, negative) [7]. These results suggest that the broader the emotion group, the better the accuracy. This new taxonomy proposal inspired us to explore different emotion categories and also confirmed that a BERT based model would be suitable for our aims. The Dataset has been used and analyzed in following studies, with Wang, K. et al. [20] achieving comparable or better results, while comparing different models and a fine-tuned BERT model. For this study, we decided to use DistilBERT [18] as our BERT based model, as DistilBERT is a streamlined version of BERT developed by a Hugging Face team that achieves significant reductions in model size and inference time of BERT while maintaining most of its performance.

The paper also examines limitations of the dataset and ways to address them, such as the big class imbalance and biases present in the dataset. We intend to expand on this notion in Section 2. To help with class imbalance, Wang, K. et al. [20] explores data augmentation methods such as Easy Data Augmentation, BERT Embeddings, and Bert-based *ProtAugment*; however, the improvement was marginal, with an increase of 0.027 from the F1 score of the original dataset. Nevertheless, they still achieved a significantly better performance on underrepresented emotion labels, which they attributed to using 10 training epochs instead of Demszky, D. et al.'s [7] 4 epochs.

Positive		Negative		Ambiguous
admiration	joy	anger	grief	confusion
amusement	love	annoyance	nervousness	curiosity
approval	optimism	disappointment	remorse	realization
caring	pride	disapproval	sadness	surprise
desire	relief	disgust		
excitement		embarrassment		
gratitude		fear		

Figure 2: Emotions Comprised in the Dataset.

Emotion Recognition in natural language processing is a complex field, with different nomenclatures, models and frameworks [16]. For this report, we use emotion recognition, emotion prediction and emotion classification interchangeably as the classification of one or multiple emotions portrayed in a specific written text. Following Plaza-del-Arco, F.M. et al.'s [16] study, we also outline that we follow the discrete model of emotions, where each emotion is distinct between each other; however, opposed to other studies in emotion classification [6], we will take advantage of the dataset's collection methodology [7] and use both the multi-label facet and human label variation available to avoid masking the degree to which annotators disagree [6]. We will elaborate this further in the sections that follow.

4 Methodology

In this section, we outline the approach and tools used to carry out our study, from data preprocessing to model training and evaluation. The entire workflow consists of Data exploration, pre-processing from data driven decisions, modeling, and evaluating. Additionally, we developed a user interface to visualize the output of these models. The entire process, including model implementations, is available in https://github.com/yanazlatanova/emotion-recognition.

4.1 Data Exploration and Preprocessing

4.1.1 *Dataset Review:*

Our initial data exploration, along with a review of the paper by the dataset creators [7], helped us better understand the dataset and informed our pre-processing strategy.

We made a graph of the distribution of the length of the comments as we can see on Figure 3. We found that the distribution was normal and that most of the comment length was around 60 characters. When we selected the comments of length inferior to 4 characters, it showed "yes/no" comments or emoticons eg. such as *:^)* and *XD*.

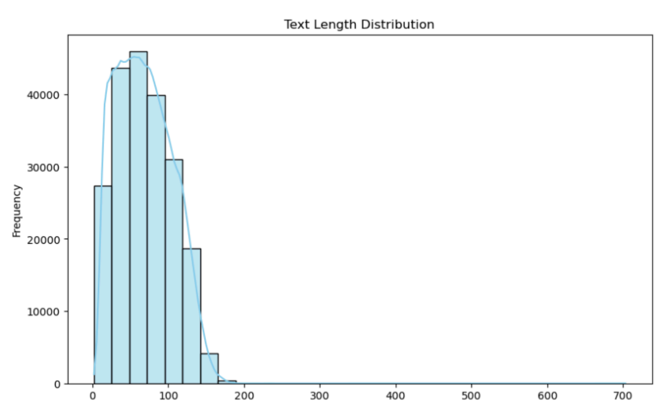

Figure 3: Comment Length Distribution

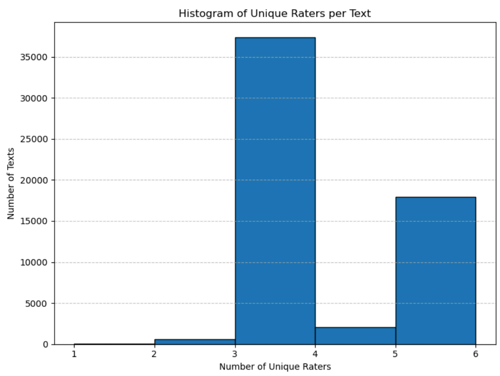

Figure 4: Number of unique raters per comment/text.

4.1.2 *Investigating Labeling Disagreement*:

We found that each comment/text had multiple duplicates, each rated by different annotators. While there were no missing values, some rows were marked as either having no assigned emotion or labeled as "Neutral", often reflecting that the emotion of the comment was unclear. According to Demszky, D. et al. [7], *"If raters were not certain about any emotion being expressed, they were asked to select Neutral. We included a checkbox for raters to indicate if an example was particularly difficult to label, in which case they could select no emotions."*

However, in some cases, raters still assigned an emotion to a comment that another rater found unclear or believed expressed no emotion. This introduced challenges in interpreting such instances.

To further analyze the multiple raters per text, we plotted the number of raters per comment, which can be seen in Figure 4. We could see that most texts were rated by three annotators, but some had only one or two.

Additionally, these raters often disagree in the emotions they assign, even in seemingly contradictory emotions. For example, the comment *"Definitely was a nonononoyes[1] for me there lol, I'm a horrible person"* got as fear, amusement, approval and disgust, from 5 different annotators. This noise, while problematic, is expected, since emotion classification is fairly subjective and there's multiple plausible answers [15], annotators' lived experiences influence their interpretations [6], and context is often necessary to access correctly the emotion (which the annotators did not have access to [7]), and domain knowledge (which most annotators might've not had, because of the "ever-evolving ethos" of Reddit's culture, both site-wide and within each subreddit [13]). On top of this, this dataset has been criticized before for its reliability [1, 4], with specific examples from Edwin Chen [1] outlying issues on comments containing profanity, sarcasm, internet style conventions, and culturally specific references. From, this we realize that our model may struggle to accurately predict emotions in future inputs, especially on inputs that contain these.

4.1.3 *Applying Labeling Disagreement Strategy*:

Based on these facts, we decided to:

- **Label unclear cases consistently**: We treated both "Neutral" and empty emotion labels as "Unclear", united under the *unclear* label, since in both situations raters were unable to confidently identify an emotion.
- **Filter by rater count:** The dataset was filtered to include only comments with at least three raters, ensuring more reliable and confident annotations. As shown in Figure 4, multiple comments had only one or two raters.
- **Soft Label Aggregation**: Since identical comments were often assigned different labels by different raters, we chose to aggregate the emotion ratings for each unique comment, embracing human label variation [15]. For instance, if the comment "this is adorable" received the following annotations:
  - **Rater 1:** [admiration, joy];
  - **Rater 2:** [admiration];
  - **Rater 3:** [amusement].

  The aggregated label distribution would be:
  - **admiration:** 0.67 (2 out of 3 raters);
  - **amusement:** 0.33 (1 out of 3 raters).
  - **joy**: 0.33 (1 out of 3 raters)

This aggregation gives more weight to emotions confirmed by multiple raters while still capturing the presence of less-agreed-upon emotions. It preserves the richness of the label diversity without resorting to semi-supervised learning. Essentially, it gives us a confidence level of the emotions per text. We found this preferable to majority vote aggregation since it preserves different annotators' perspectives [6], while avoiding the different issues that arise with assigning a "ground truth" label to a text [6, 14, 15]

While annotator disagreement and emotion uncertainty has been explored before [4, 6, 9, 21], this way of transforming the dataset seems to be a novel approach in treating annotator disagreement in both the GoEmotions dataset and emotion recognition, as it seems quite more to aggregate label disagreements into a single one [9] since it introduces uncertainty into emotion recognition, transforming our multi label classification task into multi label regression problem, using "soft" labels instead. Arguably, this also gives a hierarchical label structure to the emotions [10], enabling this dataset to be used for point-wise learning to rank tasks. While this is not our priority, our metrics and models will take this into account as well.

4.1.4 *Applying Emotion Taxonomies*:

Since emotions are complex and multidimensional, researchers have created a number of taxonomies to better classify and analyze them.

Three different emotion classification schemes were used in this study:

- Ekman's Basic Emotions taxonomy [8];
- A sentiment-based taxonomy;
- The original taxonomy of GoEmotions.

To each taxonomy, a seventh category label "unclear" was added to account for comments that are either ambiguous, emotionally neutral, or inconsistently labeled by human raters. This category helped to manage noise in the dataset, especially given the subjectivity involved in interpreting emotions from short texts.

By narrowing our classification to these seven categories, we aim to strike a balance between theoretical soundness and practical model performance, while acknowledging the limitations of emotional ambiguity in natural language.

The dataset was duplicated into three separate versions, remapping the labels according to the structure of each taxonomy. The same process outlined in Demszky, D. et al. [7] was used for the label aggregation. These versions were used in the subsequent experiments and reflected in the results.

[1] nonononoyes is a name of a subreddit dedicated to sharing videos that depict situations initially appearing to go wrong ("no, no, no...") but ultimately result in a surprisingly positive or successful outcome ("yes!").

4.1.5 *Exploring GoEmotions taxonomy:*

The GoEmotions dataset has an uneven number of comments across emotion categories, as shown in Figure 6. This makes some emotions underrepresented in the dataset, potentially resulting in a worse model performance on those categories. Because of this imbalance, our performance metrics must account for it to better evaluate how the model performs on underrepresented emotions—especially important in the context of single emotion prediction. This is also something Wang, K. et al. [20] noticed while fine-tuning a BERT model on the GoEmotions dataset, where the "grief" label, which has the least sample size in their training set, achieved the worst performance across different evaluation metrics [20].

As shown in Figure 7 some emotions are correlated as indicated by darker shades in the confusion matrix. Annoyance and anger are very much linked, as well as nervousness and fear, sadness and disappointment or joy and excitement to cite a few examples. It can be explained by the fact that some emotions are verbally implicit and need more context to be interpreted.

In their work on the GoEmotions taxonomy, Demszky, D. et al. [7] used a number of methods to investigate the consistency of emotion labeling. By using hierarchical clustering, they discovered that emotions naturally form groups based on sentiment polarity and intensity; for example, "ambiguous" emotions like surprise tended to cluster more closely with positive emotions. They used Principal Preserved Component Analysis (PPCA) to evaluate rater agreement and uncover deeper patterns, and the results indicated that all 27 emotion categories were highly distinct, which is an exceptionally strong finding in emotion research and motivated our choice of GoEmotion's taxonomy.

4.2 Modeling and Tokenization

The Hugging Face Transformers library was used to implement DistilBERT. It is well suited for emotion classification tasks due to its ability to preserve much of BERT's performance while being more efficient. Before feeding the text into the model, we used Hugging Face's tokenizer to convert sentences into token IDs. The tokenizer handles out-of-vocabulary words by breaking them into word units, ensuring even rare or misspelled terms are processed effectively.

Our models consist on fine-tuning DistilBERT to a multi-label regression task. Specifically, the text is tokenized (using DistilBert's tokenizer) and then the tokens and attention mask are fed into DistilBert's; the output of that model is fed into 2 dense layers, separated by a dropout layer, and concludes with a sigmoid activation layer over the output logits, to draw predictions per emotion[2]. To train these models, the DistilBert layers are frozen, to not only reduce computation times but also to leverage the power of the pre trained transformer. An overview of the model is shown on Figure 5.

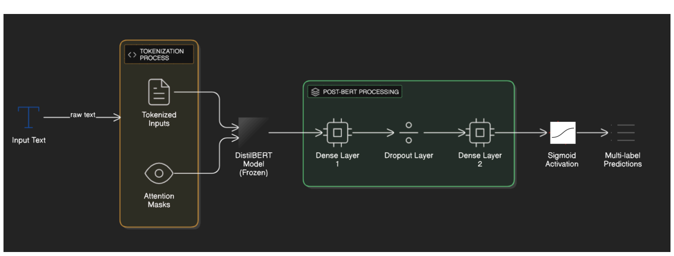

Figure 5: Model architecture diagram.

[2]We find important to note that this is preferable over a softmax activation layer because the texts can have multiple emotions.

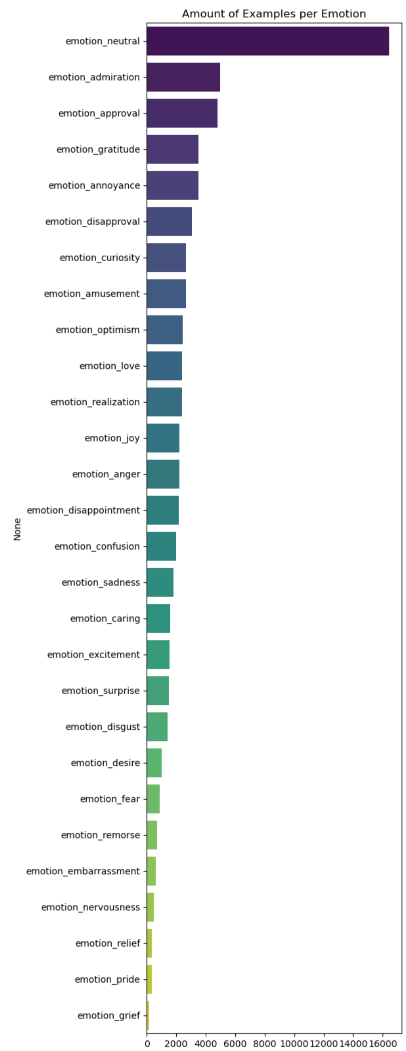

Figure 6: Distribution emotion categories.

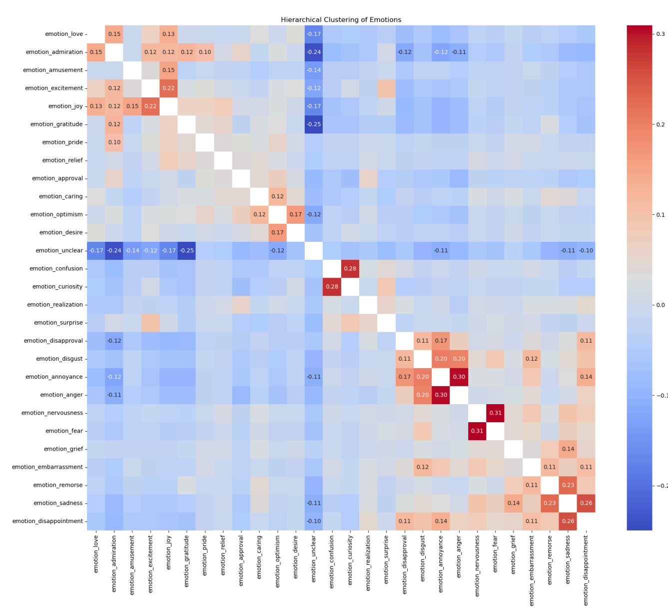

Figure 7: Confusion matrix

4.3 Training Setup

Training was conducted using PyTorch as the backend. The dataset was split into training, validation, and test sets with 70%/15%/15% ration, without any stratification, to ensure a fair assessment of the model's generalization capability.

Each model was trained for a fixed duration of 10 minutes on the same machine, using the Adam optimizer and a constant learning rate. The model was trained and evaluated using standard classification metrics that gave us information on the performance of the model, the general error and check for possible overconfidence.

4.4 Model Variants

To train our models, we decided to employ different loss functions as to optimize the modeling for different objectives. These loss functions are closely related to our performance metrics, which will be introduced better in section 5.1.

- Our first model M_{BCE} use the Mean Binary Cross Entropy (MBCE) as the loss function, to analyze the general potential of the model.
- The second model M_{DCG} uses a SoftRank-style [19] differentiable approximation of the Normalized Discounted Cumulative Gain (nDCG) for the loss function, to optimize for the expected nDCG. This model in theory should be better in ranking the emotions, while still giving relevance/confidence scores.
- The third model M_{MSE} will use the Weighted Mean Squared Error (WSME) as our loss function, giving it the power to make more bold predictions while still penalizing aggressively wrong predictions.

In parallel, every model was trained on the 3 different emotion taxonomies previously discussed, denoted in this report as M^3, M^7, and M^{28}. We generally expect for the M^3 models to have a higher performance, due to the generalized nature of the taxonomy, followed by the M^7 models for the same reason. With that being said, we still found interesting to share the performance of the M^{28} models, as the unique taxonomy associated can share more specific emotions associated with the texts beyond sentiment and the simpler taxonomies. Additionally, as we're using annotator disagreement for the prediction directly, which is fairly uncommon (and potentially novel in emotion recognition) as discussed before, the results cannot be directly compared with ones from other articles that use this dataset, due to the big difference in the annotator aggregation. Finally, we don't expect great results in general, not only due to rig and time and constraints, but also due to the dataset quality, as discussed before.

4.5 Output Visualization

Streamlit was used to create a basic user interface to present our models' predictions in an interactive way. Users can enter custom text into the interface and see the predicted emotion label and related confidence score right away. This makes it easier to understand how different emotion categories are assigned. The interface can be seen in Figure 8.

5 Evaluation

Due to the nature of uncertainty prediction and emotion complexity, it is critical that we don't focus on single metrics to evaluate the performance of our models, as that "gives no indication on how reasonable a model is, yet alone how confident and trustworthy it is" [15]. For that end, we implemented multiple different performance metrics, to measure the performance of our models in different aspects.

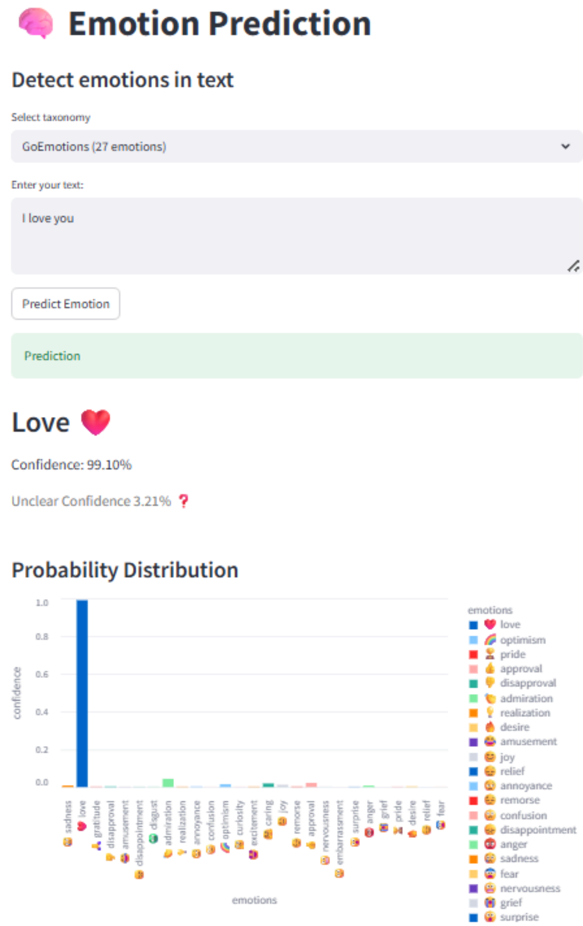

Figure 8: User interface for emotion recognition of input text.

5.1 Performance metrics

Overall, we considered and used 5 performance metrics, all of them to evaluate and compare different things. These were the following:

- Mean Binary Cross Entropy (MBCE), designed to give us information on the general error of the model, as well as its overconfidence;
- Weighted Mean Squared Error (WMSE, $w(y, \hat{y}) = e^y$), designed to check the under-confidence of the model, incentivizing it to be more bold with its predictions, while still taking into account high errors;
- Normalized Discounted Cumulative Gain (nDCG), commonly metric in information retrieval and in learning to rank tasks, to measure the model in a raking scenario, by evaluation the actual actual ranking to the ideal one, rewarding highly relevant items that appear earlier in the list, enabling us to understand if the model is correctly giving higher confidence scores to higher confidence emotions even in low confidence scenarios
- 2 macro-averaged F_1 metrics, differentiating on the true label definitions we use, designed to evaluate our model in a classification scenario, while giving us more insight on the ability of the model to predict less common emotions.

For these F_1 metrics, as the predicted labels, we decided on using 0.5 as the threshold for a positive or negative prediction; for the ground truth, for the metric F_1^{any}, we say a label is positive if any of the annotators rated as such, and for the metric F_1^{conf}, we say

Table 1
The results from our models

	3 emotions			7 emotions			28 emotions		
	$\mathrm{M_{BCE}}^3$	$\mathrm{M_{DCG}}^3$	$\mathrm{M_{MSE}}^3$	$\mathrm{M_{BCE}}^7$	$\mathrm{M_{DCG}}^7$	$\mathrm{M_{MSE}}^7$	$\mathrm{M_{BCE}}^{28}$	$\mathrm{M_{DCG}}^{28}$	$\mathrm{M_{MSE}}^{28}$
MBCE ↓	**0.489**	0.64	0.499	**0.286**	0.447	0.299	**0.1203**	0.509	0.124
F_1^{any} ↑	0.561	0.222	**0.688**	0.252	**0.403**	0.337	0.095	**0.346**	0.152
F_1^{conf} ↑	**0.511**	0.32	0.479	**0.279**	0.177	0.263	0.087	0.051	**0.103**
WMSE ↓	0.131	0.276	**0.118**	0.0653	0.11	**0.0649**	0.023	0.149	**0.022**
nDCG ↑	0.924	0.925	**0.929**	**0.89**	0.884	0.884	**0.801**	0.789	0.799

that the emotion with the most confidence, and every emotion with more than 0.8 confidence, are positive.

5.2 Results and Discussion

After training and testing the various models using the workflow described in the previous section, we obtained the results in Table 1. From them, we can make several conclusions:

- As expected, the taxonomies with more emotions had a harder time in performing better, except when comparing the WMSE metric results; this probably means that the weight chosen wasn't penalizing lower ground truths hard enough, or this low values might be a reflection of class imbalance.
- All models are generally good at ranking the emotions, as the nDCG is high across all models. While this is not surprising in the 3 and 7 emotion taxonomies, the 28 emotion taxonomy having such high values suggest that the model is great at predicting the top emotions of the comments.
- Unexpectedly, the M_{DCG} models aren't the best ones on the nDCG metric (even though they're all similar values between them), which suggests either an issue in implementation, the unsuitability of the approximated nDCG constructed, or that the model might be overfitting if optimizing for that metric. The overall performance observed in the rest of the metrics (except in the F_1^{any}) might suggest this overfitting hypothesis.
- Surprisingly, it seems that taxonomies with lower emotions have higher MBCE and WMSE than the higher emotion number taxonomy. This seems counter intuitive, but it's probably because of the big imbalance of labels per text (as most emotions should be 0). This suggests that alternative ways to model to be aware of this imbalance might be more appropriate, as well the importance of choosing a more appropriate weight function for the WMSE.
- The classification metrics pretty unstable, showing fairly different values between the different models. Regardless, from them, we can see that the models aren't as good in predicting the multiple labels directly. The M_{DCG}, weirdly enough, seems to be able to handle this better than the other emotions, when looking at the F_1^{any} metric. This might be because the model gives more confidence in the output for the emotions, as the order in the metric is more important, and there are more positive labels for the threshold we decided. This would also explain the fairly low F_1^{conf} throughout the models (including M_{DCG}), possibly suggesting that a better threshold for it should've been chosen.
- Both the M_{BCE} and M_{MSE} models perform very similarly according to the regression metrics, and both seem to be capable of predicting the emotion of the texts close to the real predictions.

Overall, the results we're in line with our expectations, and from them it seems that using and fine-tuning Distilbert is suitable for emotion recognition when taking into account annotator disagreement.

6 Conclusion

In this report, we analyzed the GoEmotions dataset with the objective of doing emotion recognition on different emotion taxonomies, these comprised of 3, 6 and 27 emotions, with the latter ones including a label for unclear. While researching and analyzing, we found some issues of the dataset, including annotator disagreement, data quality issues, and label imbalance. After aggregating the annotators based on confidence per label, we fine-tuned DistilBERT using different loss functions on those taxonomies. We could conclude that for most loss functions, the models mostly performed fairly similar between loss functions, but the different taxonomies had a huge impact on model performance. Ultimately, we were able create models that somewhat classify and predict the emotions in Reddit comments accurately, even with heavy annotator disagreement.

In future work, there are several key directions we should explore to improve both the accuracy and reliability of emotion recognition in our models. First, we should address the label noise and cultural bias in the GoEmotions dataset. As noted during our analysis, many labels appear to be inconsistent due to annotator misunderstanding of cultural references or sarcasm. A valuable improvement would be to refine or relabel parts of the dataset using more culturally diverse annotators or by adding contextual information (e.g., surrounding conversation or media references) to help raters make more accurate judgments.

Second, although we used DistilBERT for its efficiency, experimenting with larger or more specialized language models like RoBERTa, DeBERTa, or emotion-specific transformers could help us capture more nuanced emotional signals, especially for complex cases like sarcasm or mixed emotions. We could also explore prompt-tuned or instruction-following models, which might generalize better with less fine-tuning. Additionally, data augmentation techniques has shown to improve model performance and lift the burden somewhat from label imbalance [5, 11, 20].

Third, to improve the model's robustness and calibration, we should consider applying calibration techniques such as temperature scaling or isotonic regression before training, especially since our evaluation showed that while confidence scores were often reasonable, some configurations still showed signs of over- or under-confidence.

Finally, we should perform a more thorough error analysis. For instance, looking into which specific emotions are most often misclassified, and under what linguistic conditions, would help us fine-tune the model and dataset further. Another option is to use model explainability techniques like the use of shapley values [3] or LIME [17]. Future work could also expand on the hard emotion distinctions possibly incorporating label correction [12], or to incorporate fuzzy probability theory into the mix.

References

[1] 30% of Google's Emotions Dataset is Mislabeled: 2025. *https://www.surgehq.ai/blog/30-percent-of-googles-reddit-emotions-dataset-is-mislabeled*. Accessed: 2025-06-01.

[2] Ameer, I. et al. 2023. Multi-label emotion classification in texts using transfer learning. *Expert Systems with Applications*. 213, (2023), 118534. DOI:https://doi.org/https://doi.org/10.1016/j.eswa.2022.118534.

[3] An introduction to explainable AI with Shapley values: 2018. *https://shap.readthedocs.io/en/latest/example_notebooks/overviews/An%20introduction%20to%20explainable%20AI%20with%20Shapley%20values.html*. Accessed: 2025-06-01.

[4] Capstick, A. et al. 2025. Training Neural Networks on Data Sources with Unknown Reliability. (2025).

[5] Data Augmentation for Emotion Detection in Small Imbalanced Text Data: 2023. *https://arxiv.org/abs/2310.17015*.

[6] Davani, A.M. et al. 2022. Dealing with Disagreements: Looking Beyond the Majority Vote in Subjective Annotations. *Transactions of the Association for Computational Linguistics*. 10, (2022), 92–110. DOI:https://doi.org/10.1162/tacl_a_00449.

[7] Demszky, D. et al. 2020. GoEmotions: A Dataset of Fine-Grained Emotions. (2020).

[8] Ekman, P. 1992. Facial Expressions of Emotion: New Findings, New Questions. *Psychological Science*. 3, 1 (1992), 34–38. DOI:https://doi.org/10.1111/j.1467-9280.1992.tb00253.x.

[9] Fornaciari, T. et al. 2021. Beyond Black & White: Leveraging Annotator Disagreement via Soft-Label Multi-Task Learning. *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (Online, Jun. 2021), 2591–2597.

[10] Galke, L. et al. 2025. Are We Really Making Much Progress in Text Classification? A Comparative Review. (2025).

[11] Imran, M.M. et al. 2023. Data Augmentation for Improving Emotion Recognition in Software Engineering Communication. *Proceedings of the 37th IEEE/ACM International Conference on Automated Software Engineering* (Rochester, MI, USA, 2023).

[12] Jinadu, U. and Ding, Y. 2024. Noise Correction on Subjective Datasets. *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (Bangkok, Thailand, Aug. 2024), 5385–5395.

[13] Oddný, L. et al. 2023. Impact of Reddit Community Culture on User Attitude Expression and Social Interaction. *Journal of Linguistics and Communication Studies*. 2, 4 (Oct. 2023), 61–67.

[14] Ovesdotter Alm, C. 2011. Subjective Natural Language Problems: Motivations, Applications, Characterizations, and Implications. *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies* (Portland, Oregon, USA, Jun. 2011), 107–112.

[15] Plank, B. 2022. The ``Problem'' of Human Label Variation: On Ground Truth in Data, Modeling and Evaluation. *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing* (Abu Dhabi, United Arab Emirates, Dec. 2022), 10671–10682.

[16] Plaza-del-Arco, F.M. et al. 2024. Emotion Analysis in NLP: Trends, Gaps and Roadmap for Future Directions. *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)* (Torino, Italia, May 2024), 5696–5710.

[17] Ribeiro, M.T. et al. 2016. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. (2016).

[18] Sanh, V. et al. 2020. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. (2020).

[19] Taylor, M. et al. 2008. SoftRank: optimizing non-smooth rank metrics. *Proceedings of the 2008 International Conference on Web Search and Data Mining* (Palo Alto, California, USA, 2008), 77–86.

[20] Wang, K. et al. 2024. Large Language Models on Fine-grained Emotion Detection Dataset with Data Augmentation and Transfer Learning. (2024).

[21] Weerasooriya, T.C. et al. 2021. Improving Label Quality by Jointly Modeling Items and Annotators. (2021).